TURBULENCE FIDELITY

From a point aircraft to a distributed airframe

How the physics engine works, what the wind-shear model assumed, what that assumption cost, and what replaces it.

This report covers a single change and everything it rests on: the wind field used to be sampled at one point, the aircraft's centre of gravity, and its gradient taken there. For a disturbance forty wingspans across that is exact enough to be uninteresting. For a vortex core two to three wingspans across it is the weakest link in the project's headline result.

The theory backbone throughout is Stengel, Flight Dynamics, 2nd edition. Every equation used is cited by number. Two of them turned out to be wrong, which is why the verification section comes before the implementation section.

Every number here is computed, not typed

This document is generated by scripts/turbulence_report.py, which imports the model and calls it. The tail arm, the correction profile, the loading-shape spread and the ledger counts are all recomputed at build time. A regression shows up as a changed figure in the report rather than a stale claim in prose.

Boeing 747 at CR-2144 flight condition 9 - 12,192 m, 235.9 m/s, Mach 0.80

Parks et al. 1985 Kelvin-Helmholtz vortex, Hannibal case

What this document covers

Four questions, in order. Each section says which is being answered.

The rule this project runs on

A figure without its table has broken the project. That rule predates this work and is why the verification section exists: the model's three gust-rate signs were re-derived from first principles rather than transcribed, and doing so found two errors in the published source they were being checked against.

> **Provenance is now enforced, not just intended**
>
> Every constant carries a category in flightsim/provenance.py: SOURCED read from a cited table, DERIVED computed from sourced values by a stated relation, CALIBRATED fitted to reproduce a sourced number, DECLARED chosen and carrying a measured sensitivity. A test asserts that DERIVED chains name inputs that exist, are acyclic, and bottom out in something sourced. A constant added without an entry fails the build.
> Current ledger: 6 sourced, 3 derived, 1 calibrated, 3 declared.

What is deliberately not here

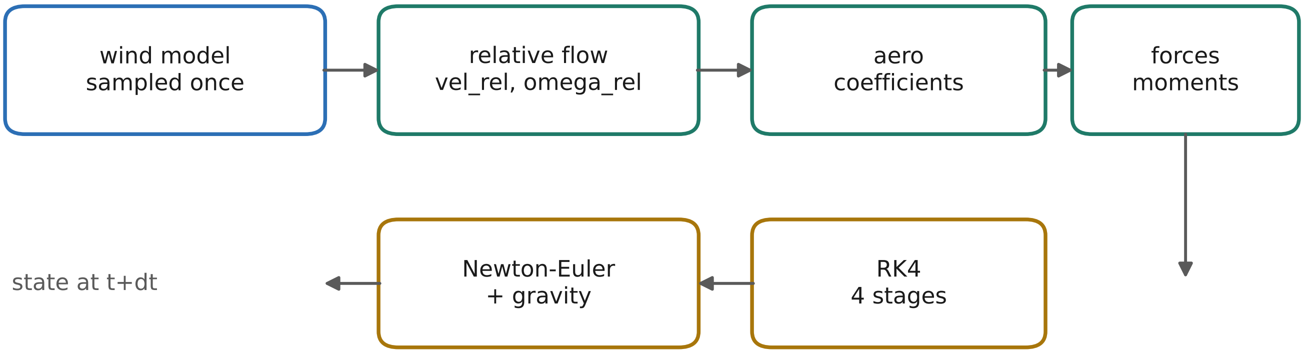
This is not a claim that the model represents a real Boeing 747. Every comparison in the project is closed-loop against a source document's own arithmetic, which is what makes it a test of the solver rather than of the aeroplane. The derivative data is labelled Flexible in its own source while this model is rigid, and that gap is unquantifiable from the documents held. Vortex conclusions stay orderings, never values.

The engine, as it stands

Thirteen state variables, integrated by fixed-step RK4 at 50 Hz. Position and attitude in an Earth frame, velocity and rotation in a body frame, with a quaternion carrying the rotation between them.

```
pos_ned  (3,)   north, east, down          quat  (4,)   w x y z, body -> NED
vel_body (3,)   u v w, body axes           omega (3,)   p q r, body axes
```

The loop, once per step



The one rule that makes turbulence tractable

The aerodynamics never see inertial velocity. dynamics.py forms vel_rel = vel_body - dcm.T @ wind_ned and omega_rel = omega - omega_gust, and passes only those to aero.py. The Coriolis, gyroscopic and kinematic terms deliberately keep the INERTIAL velocity and rate: a gust changes the flow the wings see, not the airframe's ground velocity.

Why there is no minus-m-dW-by-dt term

Stengel sets out two ways to handle a moving air mass. Work in the Earth frame and subtract the wind only where the aerodynamics need it (his eq. 3.2-85), or ride with the air mass. The second is elegant while the wind is uniform, because a block of air drifting at constant speed is itself inertial (eq. 3.2-87). The moment the wind varies in space, that frame accelerates and an apparent force -dW/dt must be added (eq. 3.2-89), where dW/dt is the material derivative: the local change plus the part you fly into (eq. 2.1-13).

That term is the price of the second choice, not a physical force waiting to be added to the first. This engine uses the first. Adding it would double-count, and the project verified that by measurement: with all aerodynamics switched off so free fall has a closed form, flying through a wind swinging at 9 g of dW/dt matched the exact solution to 4e-12 m. Injecting the term deliberately gave 13.33 m.

Where the Earth frame becomes the body frame

The weather is described in North-East-Down. It has to be: a vortex sits at a fixed place in the sky and does not care where the aeroplane points. The aerodynamics only work in body axes, because angle of attack means how far the nose is tilted relative to the oncoming air. Somewhere the two must meet, and in this engine that happens at exactly two lines.

The wind vector: one rotation

```
vel_rel = vel_body - dcm.T @ wind_ned          Stengel eq. 3.2-85
V, alpha, beta from vel_rel                    Stengel eq. 3.2-86
```

The wind gradient: a rotation on both sides

```
jac_ned   = jacobian(field)(pos_ned)           Stengel eq. 2.1-11 / 3.4-46
grad_body = dcm.T @ jac_ned @ dcm             Stengel eq. 3.4-47
```

A gradient answers the question how much does THIS change if I step THAT way. Both halves are directional. One rotation re-expresses the answer, a wind component; the other re-expresses the question, a direction to step in. Rotating one side only would leave a quantity half in each frame. Stengel names this a similarity transformation and the code performs exactly it.

> **The contract is deliberately asymmetric**
>
> wind_model(wind_state, state, key, dt) -> (wind_ned, omega_gust, wind_state, key)
> wind_ned crosses in NED. omega_gust crosses ALREADY IN BODY AXES, because it is subtracted from state.omega, which is body. Above the handover everything is NED and attitude-blind; below it everything is body and never sees NED.

What changes as the aircraft moves through a field

What	Consequence
Position moves	the field is re-evaluated; new wind and new gradient
Attitude changes	the same air maps to different body components
Time	nothing: the field is frozen in the Earth frame

The second is the one that surprises. Freeze the aeroplane in space and roll it ninety degrees: the wind in NED is bit-identical, yet vel_rel and omega_gust both change. For the Parks vortex this has a sharp consequence. The vortex axes lie across the flight path and the field has no lateral variation at all, so flying wings-level due north the roll and yaw gust inputs are exactly zero and the aircraft is pitched and nothing else. Roll appears only once it banks, created purely by the rotation of axes rather than by meeting different air.

Why a gust gradient is a rotation

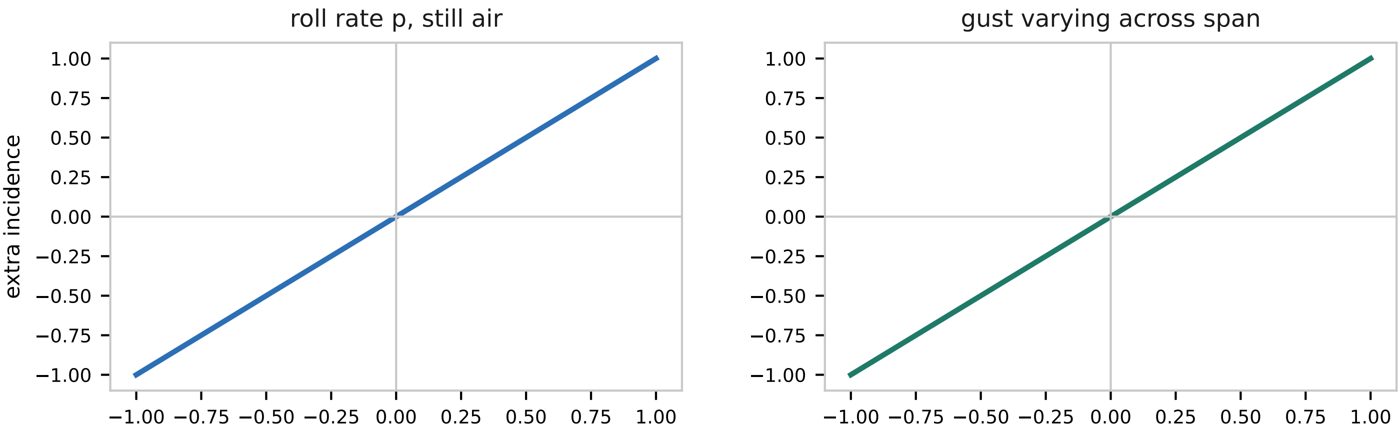
This is the idea the whole turbulence model rests on, and it is worth stating carefully because it is what licenses treating a spatially varying wind with derivatives that were measured for a rigidly rotating aircraft.

Start with the aircraft rolling in still air

At roll rate p, the wing at spanwise station y moves downward at p*y. It meets the air from slightly below, so it sees an extra angle of attack. The further outboard, the larger the effect, and the tips trace helices.

```
d(alpha) = p*y / V                          Stengel eq. 3.4-39
```

That linearly varying incidence, integrated across the span, IS the roll damping derivative Clp. Stengel's Fig. 3.4-3 draws the resulting spanwise lift distribution.



Now hold the aeroplane still and let the AIR have a vertical velocity that varies across the span. The wing sees the same picture. It cannot tell the two situations apart, and neither can any derivative measured on it.

The six equivalences

```
d(w)/dy = +p    d(v)/dz = -p       (3.4-48, 3.4-49*)
d(w)/dx = -q    d(u)/dz = +q       (3.4-50, 3.4-51)
d(v)/dx = +r    d(u)/dy = -r       (3.4-52, 3.4-53)
```

The model uses the left-hand column. The asterisk on 3.4-49 is not a footnote marker; the sign printed in the source is wrong, and the next page explains how that was established.

Verifying the source before building on it

The three signs the model uses were re-derived rather than transcribed, by expanding the relative flow at a body-fixed point and matching coefficients.

```
v_rel(r) = v_cg + omega x r - w_g(r)
omega x r = (q*z - r*y,  r*x - p*z,  p*y - q*x)
```

Each composite formula was then given a rigid-rotation self-consistency test: substitute a shear matrix that exactly matches a rigid rotation, and check the moment returned equals the equivalent-rate moment. Thirteen formulations were checked. Eleven passed. Two did not.

> **Eq. 3.4-49 has a sign error as printed**
>
> The source gives d(v)/dz = +p. It is -p. For omega = (p,0,0) at a point directly below the centre of gravity, omega x r = (0, -p*z, 0): roll right-wing-down and the belly swings LEFT. Eqs. 3.4-54 and 3.4-56 self-check only once that sign is corrected, which is good evidence the slip is confined to that one line rather than being a different convention.

> **Eq. 3.4-55 fails by a factor of -2 and is not used**
>
> Substituting a rigid pitch rate gives d(w)/dx = -q and d(u)/dz = +q, so the bracket (d(w)/dx - d(u)/dz) evaluates to -2q and the formula returns +2*Mq*q where it should return -Mq*q. A factor of one half repairs the magnitude but cannot be justified: the tail-incidence mechanism dominates Mq for a conventional aircraft, while d(u)/dz mostly alters the tail's dynamic pressure rather than its incidence, so a 50/50 split is not physical either.

Why this changed the design

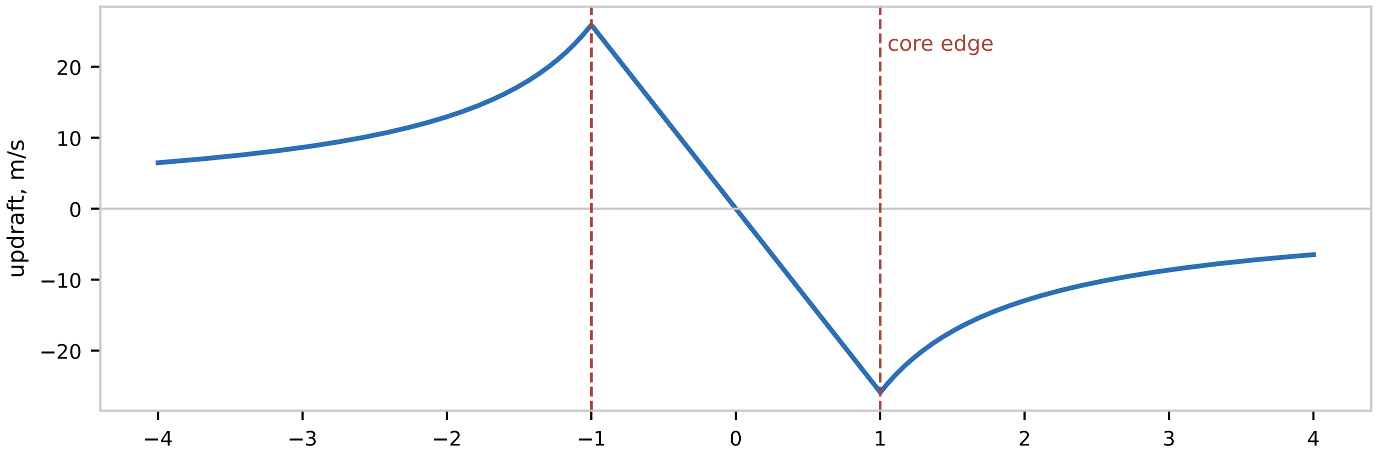
Eqs. 3.4-54 to 3.4-56 are the source's per-surface treatment: they split the roll, pitch and yaw derivatives into wing, tailplane and fin parts and drive each with the shear component that actually reaches it. That is the natural way to improve a point model, and it was ruled out twice over. The pitch equation does not survive its own self-check, and the split needs tail areas and arms that neither CR-2144 nor the Boeing simulation data it cites tabulates.

So the design took the route that never forms an equivalent rate at all: sample the field across the airframe and integrate. That is what Stengel names as required below rotor scale, on p. 217.

The disturbance being flown through

Parks et al. 1985 identified Kelvin-Helmholtz vortex arrays from two DC-10 flight-recorder traces near the tropopause. The model is a Rankine vortex: a solid-body core embedded in an irrotational outer flow, with arrays built by linear superposition, which is what makes summing it with other components legitimate.

```
inside  (r <  r0):   w_xy = V0*d/r0        w_z = -V0*l/r0
outside (r >= r0):   w_xy = V0*r0*d/r^2    w_z = -V0*r0*l/r^2
```



Through the core the vertical gust is an antisymmetric up-then-down doublet, linear inside the core and falling as 1/r outside. That antisymmetry is what separates a vortex from a single-signed gust, and it is why a vortex cannot produce a signed load asymmetry of its own.

Parks case	core r0	peak V0	spacing	r0 in spans
Hannibal, 37,000 ft	182.9 m	25.91 m/s	1067 m	3.07
Morton, 39,000 ft	137.2 m	21.34 m/s	975 m	2.30

The last column is the problem. The 747's span is 59.64 m, so the core is only two to three wingspans across. Every other field in the project is forty spans or more.

What the point assumption cost

Stengel states the validity condition for the rotary-derivative treatment plainly on p. 215: it is for shear whose length scale is large compared with the aircraft, and it assumes the wind vector and shear matrix are taken at the centre of mass with the aircraft not disturbing the flow. On p. 217 he adds the warning that matters here: stability derivatives work for large atmospheric rotors but have limited value for wake vortices, where strip theory or CFD is required.

Field	Scale	In 747 spans
Parks Hannibal core	183 m	3.07
Parks Morton core	137 m	2.30
Wingrove updraft radius	2,359 m	39.6
Doyle lee wave, quarter wavelength	6,250 m	104.8

The vortex is marginal by a factor of about thirteen

It is on the right side of Stengel's line in KIND -- these are atmospheric Kelvin-Helmholtz rotors, not aircraft wakes, so one-way coupling is safe and the derivative treatment is the right family of method. It is marginal in SIZE. Everything else in the project is comfortably a point; this is not.

What a point sample keeps, and what it loses

The model was never a naive point model. Expand the body-axis gust about the centre of gravity and three terms appear. The constant becomes the translational gust and feeds angle of attack and sideslip. The linear term becomes the rotational gust and feeds the rate derivatives. The quadratic term has nowhere to go at all.

```
w_g(r) = w_g(0)  +  (dw/dx)*x + (dw/dy)*y  +  O(r^2)
             |                 |                  |
         wind_ned        omega_gust          nothing
```

A rigid-body coefficient build-up has no channel for curvature: there is no derivative meaning the spanwise gust profile is bent. Representing it means abandoning the point-plus-gradient idea and integrating across the span, which is what the strip treatment does.

Change one: fit instead of differentiate

The first change keeps the three quantities the aero model already consumes and improves the estimator for them. Rather than the tangent at the centre of gravity, take the least-squares slope across the extent the aerodynamics actually integrate over: the span laterally, and the tail arm longitudinally.

> **The safety property that made this adoptable**
>
> A least-squares slope through samples of a LINEAR function is exactly its slope. So for any field the old model handled correctly, the new one returns the identical answer, and every existing result is provably unmoved. It differs only where the field is curved across the airframe -- which is the entire point.

The one quantity this needed, and where it came from

The lateral extent is the span, which is tabulated. The longitudinal extent is a tail arm, which is not tabulated for any aircraft the project holds. It is recovered from two derivatives that are, by taking the ratio of Stengel eqs. 3.4-7 and 3.4-12, in which the unknown tail lift slope cancels.

```
CL_qhat = 2 * CL_a,ht * (l/c)        eq. 3.4-7 non-dimensionalised
Cm_qhat = -2 * (l/c)^2 * CL_a,ht     eq. 3.4-12
------------------------------------------------------
l/c = -Cm_qhat / CL_qhat
```

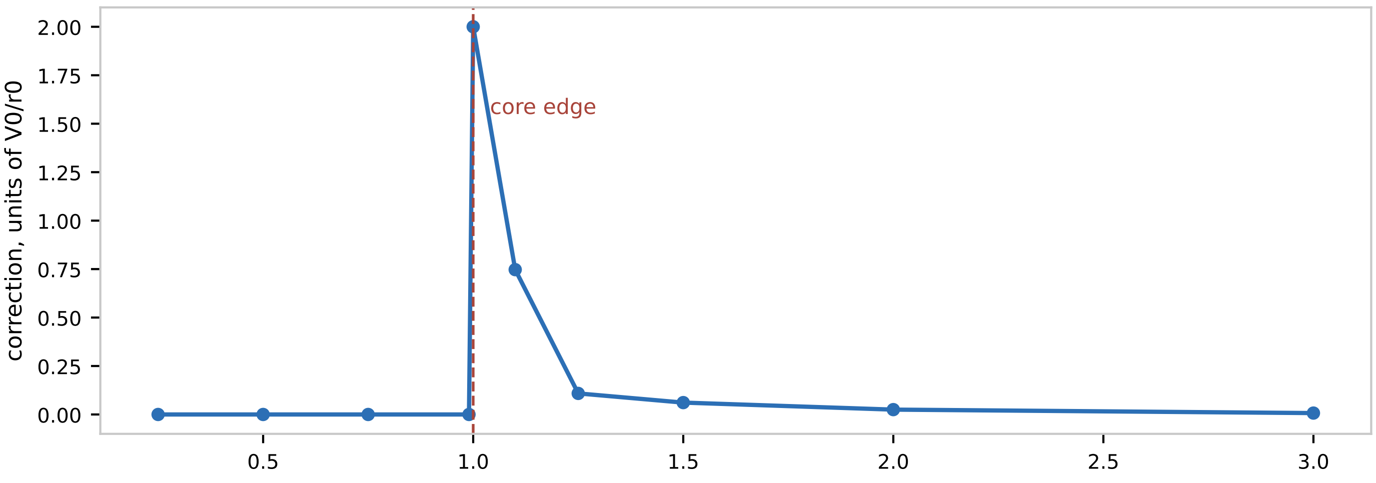
Aircraft	l/c	arm	gate
boeing747	4.0241	109.90 ft	pass
boeing747_approach	3.8519	105.19 ft	pass
cessna172	0.8558	4.19 ft	REJECTED
cherokee	1.2802	6.72 ft	REJECTED

The 747 value is 4.0241 chords, 109.90 ft. The real 747-100's centre of gravity sits roughly 100 to 110 ft ahead of the tailplane, so this lands inside the real geometry having been told nothing about it beyond the mean chord. That agreement is the entire justification for the relation, so it is asserted as a test rather than described in a comment.

The same check rejects both light-aircraft sets, whose CLq and Cmq imply arms far shorter than those airframes have. Whether the fault is the source data or the tail-dominated reading is not resolved; either way the strip path must not be used for them, and the gate enforces that.

What the point assumption actually cost

The assumption register has carried the scale ratio since session 11 but never a measured consequence. It has one now, taken along a traverse of the Hannibal core and normalised by V0/r0, the core's own characteristic pitch-rate input.



Station	Correction
0.50 r0	0.0000
0.99 r0	0.0000
1.00 r0	2.0000
1.10 r0	0.7472
1.25 r0	0.1086
2.00 r0	0.0250
3.00 r0	0.0072

Inside the core the correction is exactly zero, not merely small

Parks' Rankine profile is LINEAR in radius, so a point sample plus an analytic gradient is not an approximation at all while the airframe is inside the core. This is why the existing vortex results survived the assumption as long as they did -- and it is specific to this field. It would not hold for a Dryden field or a wake vortex.

At the boundary the gradient is discontinuous, not merely steep

Velocity is continuous at r0 -- both branches agree, which an existing test already asserted -- but the derivative is not. Inside it is +V0/r0 = +0.14167 rad/s; outside at r0 it is +0.14167. The two one-sided derivatives differ by 2*V0/r0 and have OPPOSITE SIGNS. So the tangent there is not merely inaccurate, it is ambiguous, and the code's strict less-than resolves the tie toward the outside branch -- the wrong side, since the airframe is still almost entirely inside the core. The fit has no such ambiguity.

Change two: integrate across the span

Fitting a slope still collapses the field to three numbers, so a profile that bends rather than ramps is still unrepresented. Only giving each strip the gust at its own station carries that.

```
Cl = -(1/(S*b)) * integral( y * c(y) * a0 * d(alpha)(y) dy )
```

The shape is declared; the magnitude is calibrated

The integral needs a spanwise lift distribution. Taper ratio is not tabulated in CR-2144, is not in the Boeing simulation data that report itself cites, and is not recoverable from the reference geometry: a straight-tapered wing requires a shape factor of 0.72873 from these values, and that function has a global minimum of 0.75. No real taper ratio satisfies it, because the 747 planform is cranked.

So the shape is elliptic, needing no taper ratio at all, and DECLARED. Its magnitude is pinned so that integrating a rigid roll rate reproduces the tabulated Clp exactly, which makes it CALIBRATED against a sourced number.

```
elliptic loading  =>  Clp_hat = -a0/8   =>   a0 = -8*Clp = 2.8014
```

That value looks low against a thin-airfoil 6.28 and the aircraft's own CLa of 4.94, and it should. It is an EFFECTIVE slope absorbing sweep, the tail's share of Clp, and the gap between elliptic strip theory and a real cranked swept wing. A value near 6.28 would mean the calibration had not absorbed those and would be the surprising outcome.

Shape	Rolling moment, cubic gust
elliptic	+1.9681e-06
uniform	+3.1493e-06
tapered	+2.0188e-06

The two shapes that actually taper toward the tips agree to 2.6%. The full spread of 49.7% is driven entirely by the uniform distribution, which loads the tips where a cubic gust is largest and is not a defensible transport planform. It is in the sweep as a bracket, not as a candidate.

A sign error, and how it was caught

The gust incidence was implemented as +w_g/V. It is -w_g/V. A wing moving DOWN meets the air from below and gains incidence; air moving DOWN past a stationary wing arrives from above and loses it. The rigid-rotation check could not catch this, because that path never touches a gust and reproduced the sourced Clp perfectly with the sign inverted. It took comparing the two treatments against each other on a linear gradient -- where they must agree -- to expose it. A model with that bug rolls correctly for its own motion and backwards for every gust.

Which numbers are bulletproof

Review asked for an unambiguous separation between numbers read from cited tables and numbers that were predicted, with no credit given to a predicted number for landing in a plausible range. That is now data, and a test enforces it.

Category	n	Meaning
SOURCED	6	read from a cited table, with document and page
DERIVED	3	computed from sourced values by a stated relation
CALIBRATED	1	fitted to reproduce a sourced number
DECLARED	3	chosen; must carry a measured sensitivity

The enforcing test asserts that every entry uses one of the four categories, that SOURCED and DECLARED entries carry a usable detail, that DERIVED and CALIBRATED entries name inputs which themselves exist in the ledger, and that the dependency graph is acyclic so every chain bottoms out in something sourced.

The two entries that most need reading

b747.l_eff is DERIVED and must never be quoted as 747 geometry. It attributes both pitch derivatives to the tail, and the obvious correction -- subtracting the wing's share via eqs. 3.4-13/3.4-14 with the sourced c.g. at 0.25 chord -- was tried and REJECTED: it makes the wing 83 percent of CLq, implying a 23-chord tail arm. Those are two-dimensional infinite-aspect-ratio results and they do not transfer.
strip.loading_shape is DECLARED. Quote 2.6 percent as the shape cost and 49.7 percent as the bound.

What the source audit established

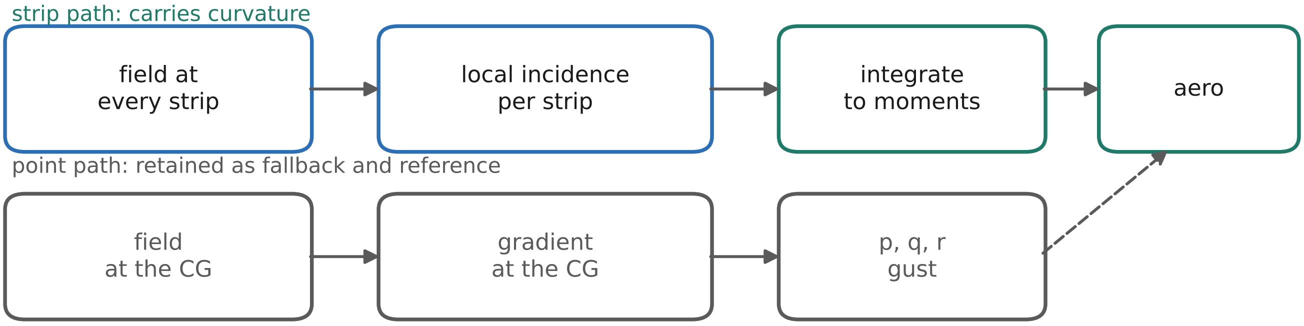
CR-2144 was read page by page, and so was the Boeing simulation report it names as its own 747 source. Neither tabulates taper ratio or any tail dimension. Both model the whole aircraft, so neither needs component geometry, and no further digging in that lineage will produce it. The audit did return three quantities the design had assumed absent: the centre of gravity at 0.25 chord, the mean chord's spanwise station, and the pilot station offset -- the last of which unblocks the accelerometer lever-arm correction that had been recorded as impossible.

How the engine will work after the next change

The strip integration is built and validated, but nothing consumes it. The 6-DOF still receives its rotational gust through the existing contract, so today the strip result is a measurement of the error rather than a correction to it. The next change wires it in.

Now

After



The two paths coexist. The point path stays the default and the reference implementation, because it is what every frozen baseline was measured against, and it remains the fallback for aircraft whose derivative sets fail the tail-arm gate. The strip path becomes opt-in per run.

> **What has to change, and what deliberately does not**
>
> The wind-model contract widens to carry a coefficient increment alongside the gust rates. dynamics.derivatives adds that increment at the same seam where wind already enters, so aero.py is untouched and keeps its rule of never seeing inertial velocity. integrate.step threads it through. The default remains zeros, so a run that does not ask for strip loads is bit-identical to today.

This will move the vortex figure-8 point. That movement is expected, is explained by the correction profile on page 10, and must be reported rather than tuned away. The ordering claim -- vortex below updraft below manoeuvre -- is what has to survive, because the project's own rule already restricts vortex conclusions to orderings.

What is still not modelled

Stated so no reader over-reads the result. None of these is an oversight; each is recorded in the assumption register with its reasoning.

Limitation	What it means
Strip theory is not CFD	each strip is treated two-dimensionally, with no spanwise induced coupling between str
The aircraft stays transparent	its own flow field does not distort the impinging vortex; genuine wake encounters becor
The loading shape is declared	the integral is right; the shape is an assumption, costing 2.6 percent across defensible a
l_eff folds wing and fuselage in	justified empirically by landing on the real tail arm, not by a computed error bar
Quasi-steady aerodynamics	no alpha-dot term. Stengel eq. 3.4-58 gives the unsteady contribution; the model has n
Constant derivatives	the same numbers at every Mach and incidence the sim reaches; unbounded, and the l
Rigid airframe	the derivative data is labelled Flexible in its own source; unquantifiable from document

Two of these get WORSE in the direction this work is aimed

The missing alpha-dot term and the rigid-airframe assumption both bite harder as fields get smaller. A shorter traverse makes unsteady lag a larger fraction of the encounter, and the sharper spanwise loads this work newly resolves are exactly the ones that would flex a real wing. Making the aerodynamic sampling finer does not make the structural model better, and it should not be read as doing so.

The honest summary is that this work removes one specific, measured error -- treating a two-to-three-span disturbance as though it acted at a point -- and leaves the others exactly where they were. That is worth doing because it was the weakest link in the project's headline result, and because it is now bounded rather than argued.

Sources

Theory

R. F. Stengel, Flight Dynamics, 2nd ed., Princeton University Press, 2022. Sections 2.1, 2.2, 3.2 and 3.4. Equations used: 2.1-10 and 2.1-11 wind field and shear matrix; 2.1-13 material derivative; 3.2-85 and 3.2-86 air-relative velocity and angles; 3.2-87 and 3.2-89 inertial versus air-mass frames; 3.2-119 acceleration at an offset point; 3.4-7 and 3.4-12 tail pitch derivatives; 3.4-39 roll-rate incidence; 3.4-40 strip-theory roll damping; 3.4-46 and 3.4-47 shear matrix and its similarity transform; 3.4-48 to 3.4-53 rate equivalences; 3.4-54 to 3.4-56 per-surface shear moments, NOT USED; 3.4-58 unsteady term, not modelled.

Aircraft data

R. K. Heffley and W. F. Jewell, Aircraft Handling Qualities Data, NASA CR-2144, December 1972, Section IX. Table IX-3 reference geometry and flight conditions, printed p. 229; Table IX-4 longitudinal dimensional derivatives, p. 230; Table IX-8 lateral dimensional derivatives, p. 234; Figure IX-1 flight conditions, p. 212; Figure IX-2 general arrangement, p. 213. Tables IX-3, IX-4 and IX-8 were verified element by element against the transcription in flightsim/aircraft.py.

C. R. Hanke and D. R. Nordwall, The Simulation of a Jumbo Jet Transport Aircraft, Volume II: Modeling Data, Boeing D6-30643-VOL-2 / NASA CR-114494, September 1970. The document CR-2144 names as its sole 747 source. Its Summary of Areas and Dimensions supplies wing area, mean chord, span, wheel geometry and engine moment arms, and no tail geometry.

Atmospheric fields

E. K. Parks, R. C. Wingrove, R. E. Bach and R. S. Mehta, Identification of Vortex-Induced Clear Air Turbulence Using Airline Flight Records, J. Aircraft 22(2), February 1985, pp. 124-129. The Rankine vortex model, eqs. (3) to (6), and the two identified cases.

R. C. Wingrove and R. E. Bach, Severe Turbulence and Maneuvering from Airline Flight Records, J. Aircraft 31(4), 1994. The updraft magnitudes and the Fig. 8 discriminator the project's turbulence work is aimed at.

Project documents

docs/PROJECT.md section 4 for measured checks and frozen baselines. docs/ASSUMPTIONS.md section E2 for the point-aircraft assumption and its bound. docs/superpowers/specs/2026-08-14-wind-shear-fidelity-design.md for the source verification and the design reasoning, including the two errors found in Stengel and the self-consistency test that found them.

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.venv/Scripts/python.exe scripts/turbulence_report.py
docs/summary/turbulence-report.pdf
Every figure and number above is recomputed on each build by calling the model.